

Pratik Kumar Padhi

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Computer Science & AI/ML · Data Analytics · Applied ML Engineering · Backend & Data Engineering

Summary

Final-year B.Tech in Computer Science & Engineering (AI & ML), building data and ML systems end to end: ingestion, model training, REST APIs and deployment. Two of the three projects below are deployed, each with a test suite or validation script behind it. Open to SDE, ML engineering, data analyst and backend roles from May 2027.

Education

Centurion University of Technology and Management

B.Tech in Computer Science & Engineering – AI & Machine Learning Specialization

Bhubaneswar, Odisha

Aug 2023 – May 2027

Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Database Management Systems, Computer Vision, Statistics for Data Science

Technical Skills

Languages: Python, SQL, JavaScript, C++, Java

Data & Analytics: Pandas, NumPy, Power BI, DAX, Power Query, Data Modeling, Statistical Analysis, Matplotlib, Seaborn, Jupyter

Machine Learning & AI: scikit-learn, Stacking Ensembles, Model Evaluation & Cross-Validation, TensorFlow, PyTorch, Keras, OpenCV, Computer Vision

Backend & APIs: Flask, FastAPI, REST API Design, SQLite, Gunicorn, pytest, Unit Testing, Integration Testing, Data Pipelines

Cloud & Tools: AWS, Docker, Linux, Git, GitHub, Render, Netlify, Postman

Enterprise & SAP: SAP S/4HANA, ABAP (ABAP Cloud), SAP Fiori, Business Process Integration

Projects

LedgerLoop – Multi-Source Payment Reconciliation & Settlement Engine

Sep 2026

<https://github.com/Pratik-padhi/LedgerLoop-multi-source-payment-reconciliation-and-settlement-engine>

Python • Flask • SQLite • pytest • Gunicorn • Gemini • HTML/CSS/JavaScript • Render

- Built a schema-driven engine that reconciles payment gateway, bank and internal ledger records through four escalating tiers, from a deterministic exact match up to guarded LLM adjudication.
- Backed it with **415 automated tests** across 15 files, covering normalization, Decimal accounting and the fallback path of every matching tier. The suite runs offline with the LLM provider switched off, so a rerun is deterministic.
- Correctness mattered more than throughput here. Settlement arithmetic uses Decimal, every upload is fingerprinted with SHA-256 for lineage, and each LLM verdict is validated server-side before it is accepted. Anything ambiguous routes to human review.
- Shipped a Flask controller UI over 7 REST endpoints, with SQLite persistence and a Gunicorn deployment on Render.

CropSensy – Microclimate Crop Detection & Agronomy Advisory System

Apr – Aug 2026

https://github.com/Pratik-padhi/Cropsensy-MicroClimate_Crop-detection

Python • scikit-learn • Flask • Pandas • NumPy • joblib • JavaScript • Leaflet • Netlify • Render

- Trained a stacking ensemble – Random Forest (100 trees), Decision Tree and a Logistic Regression meta-learner – to classify 4 crops, reaching **97.25% test accuracy** and $97.10\% \pm 0.36\%$ mean 5-fold cross-validation accuracy.
- Engineered 27 features from 7 soil and microclimate inputs the farmer types in. A 327-line `validate.py` rejects out-of-domain values and exits non-zero if any get through.
- Seven Flask endpoints cover prediction, soil diagnosis and fertilizer recommendation. The frontend fills in location and live weather from Open-Meteo.
- Model weights ship as joblib artifacts, so training and serving are decoupled. Frontend on Netlify, backend on Render.

Uber NCR Ride Analytics – Power BI Revenue & Operations Dashboard

Jun – Aug 2026

https://github.com/Pratik-padhi/Uber_Analytics_Dashboard_using_PowerBI

Power BI • DAX • Power Query • Data Modeling • Bing Maps

- Built a 5-page Power BI report over **150,000 Delhi NCR bookings**, surfacing INR 51.85M revenue, a 62% completion rate, INR 4.59M lost revenue and 176 pickup locations.
- Power Query transforms the source data into the model; DAX measures on top of it isolate peak booking windows, the drivers behind cancellations, and vehicle-level trends across 7 vehicle types.
- Bing Maps ranks pickup zones by volume and completion rate.

Research & Presentations

ICCAIML 2026 – Co-Authoring Conference Paper, Hyderabad, India

2026

“Transpiration Crop Yield Impact and Intelligent Weed Density Analysis Via Artificial Intelligence”

Published in *Information System and Data Analytics*, ed. Subhendu Pani. CRC Press, Taylor & Francis Group.

Certifications

SAP Certified Associate – Business Process Integration with SAP S/4HANA

2025

SAP Certified – Back-End Developer, ABAP Cloud

2025

SAP Certified – SAP Fiori Application Developer

2025